Tsirelson bounds for quantum correlations with indefinite causal order

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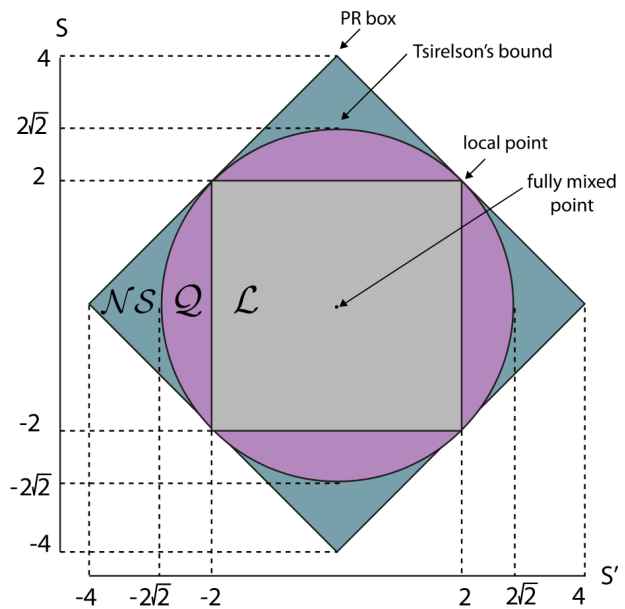
Tsirelson bounds in Bell nonlocality

- Quantum correlations between distant events:

$$p(ab|xy) = \langle\psi|M_{a|x}N_{b|y}|\psi\rangle$$

- **Tsirelson's bound**: maximal quantum CHSH correlation $2\sqrt{2}$ (classical bound: 2).
- General method: NPA hierarchy



N. Brunner, D. Cavalcanti, S. Pironio, V. Scarani, and S. Wehner (2014)

This talk is about ...

Bell-like inequalities for causality

deriving the limit of quantum correlations with indefinite causal order

Outline

1. What is a causal inequality?
2. Quantum operations with indefinite causal order (ICO)
3. **Tsirelson bounds for correlations with ICO**
4. Open questions and outlook

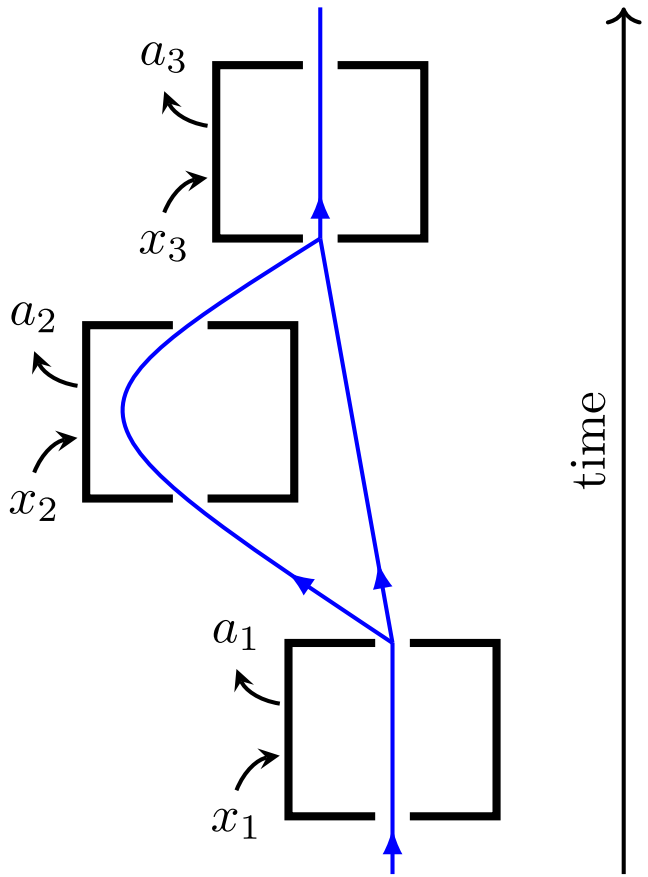
What is a causal inequality?

Assumptions

- free choice
- closed laboratories
- global causal structure

observed statistics (**causal probability distribution**):

$p(a_1, \cdots, a_N \mid x_1, \cdots, x_N)$



O. Oreshkov, F. Costa & Č. Brukner (2012)

Bell-like inequalities

Causal inequalities

constraints for causal probability distributions:

$$p(\vec{a} \mid \vec{x}) \text{ is causal} \Longrightarrow \mathcal{J} \leq \beta$$

- settings $\vec{x} = (x_1, \cdots, x_N)$
- outcomes $\vec{a} = (a_1, \cdots, a_N)$
- real coefficients $\left(\alpha_{\vec{a},\vec{x}}\right)$
- an N-party correlation

$$\mathcal{J} = \sum_{\vec{a},\vec{x}} \alpha_{\vec{a},\vec{x}}\, p(\vec{a} \mid \vec{x})$$

O. Oreshkov, F. Costa & Č. Brukner (2012)

The set of causal probability distributions

- Two-party causal probability distribution: random mixture of
  $B \not\leftarrow A \quad p_A(a_1 \mid x_1, x_2) = p_A(a_1 \mid x_1, x_2'), \forall x_1, x_2, x_2', a_1$
  $A \not\leftarrow B \quad p_B(a_2 \mid x_1, x_2) = p_B(a_2 \mid x_1', x_2), \forall x_1, x_2, x_1', a_2.$
- Causal probability distributions form a **convex polytope** in $\mathbb{R}^{\prod_i m_i n_i}$, m_i (n_i) being the number of outcomes (settings) of the i-th party.

C. Branciard, M. Araújo, A. Feix, F. Costa and Č Brukner (2016)
O. Oreshkov and C. Giarmatzi (2016)
AA. Abbott, C. Giarmatzi, F. Costa, and C. Branciard (2016)

Facets of the simplest causal polytope

- Guess Your Neighbor's Input (GYNI), uniform settings

$$\mathcal{J}_{\text{GYNI}} := \Pr(a_1 = x_2, a_2 = x_1)$$

- Lazy Guess Your Neighbor's Input (LGYNI), uniform settings

$$\mathcal{J}_{\text{LGYNI}} := \Pr(x_1(a_1 \oplus x_2) = 0, x_2(a_2 \oplus x_1) = 0)$$

$$\mathcal{J}_{\text{GYNI}} \leq \tfrac{1}{2}, \mathcal{J}_{\text{LGYNI}} \leq \tfrac{3}{4}.$$

C. Branciard, M. Araújo, A. Feix, F. Costa and Č Brukner (2016)

Oreshkov-Costa-Brukner (OCB) inequality

- Alice's setting: a uniformly random bit x_1,
- Bob's setting: a pair of uniformly random bits (b, c).

$$\mathcal{J}_{\mathrm{OCB}} := \Pr(a_1 = b \mid c = 0) + \Pr(a_2 = x_1 \mid c = 1)$$

OCB inequality: $\mathcal{J}_{\mathrm{OCB}} \leq \frac{3}{2}$.

O. Oreshkov, F. Costa & Č. Brukner (2012)

Violations of causal inequalities

Examples

- $\mathcal{J}_{\text{GYNI}} = 0.6218 > \frac{1}{2}$;
- $\mathcal{J}_{\text{LGYNI}} = 0.8194 > \frac{3}{4}$;
- $\mathcal{J}_{\text{OCB}} = 1 + \frac{1}{\sqrt{2}} > \frac{3}{2}$;

Quantum theory is compatible with processes with **indefinite causal order**.

What are the Tsirelson-like bounds for causal inequalities?

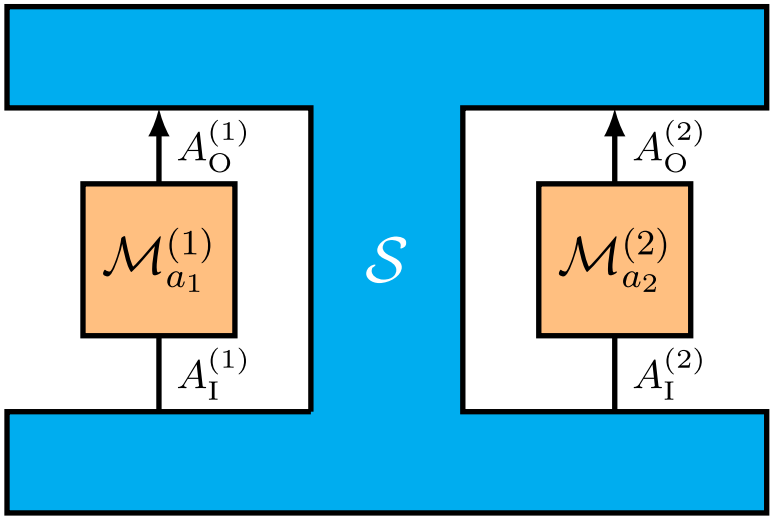
O. Oreshkov, F. Costa & Č. Brukner (2012)
C. Branciard, M. Araújo, A. Feix, F. Costa and Č Brukner (2016)

Quantum operations with indefinite causal order (ICO)

Local quantum theory and correlations with ICO

- drop the assumption of a global causal structure;
- keep logical consistency with local quantum operations:

$$p(\vec{a}) = \mathcal{S}\left(\mathcal{M}^{(1)}_{a_1}, \cdots, \mathcal{M}^{(N)}_{a_N}\right)$$



G. Chiribella, G. D'Ariano, P. Perinotti, and B. Valiron (2009)
O. Oreshkov, F. Costa & Č. Brukner (2012)

Process matrices

Choi-Jamiołkowski isomorphism:

$$\mathcal{M}_{a_i}^{(i)} \Longleftrightarrow M_{a_i}^{(i)}$$

$$\mathcal{S} \Longleftrightarrow S \quad \text{(process matrix)}$$

Born rule in local quantum theory:

$$p(\vec{a}) = \mathrm{Tr}\left[S^{\mathrm{T}} \bigotimes_{i=1}^{N} M_{a_i}^{(i)} \right]$$

- S is a postive operator;
- S is in the dual affine set of (Choi operators of) no-signalling channels.

O. Oreshkov, F. Costa & Č. Brukner (2012)
G. Chiribella and D. Ebler (2016)

Previous results involving the boundary of ICO

- Brukner (2015): $\mathcal{J}_{\mathrm{OCB}} = 1 + \frac{1}{\sqrt{2}}$ is the maximum with a restricted set of local quantum operations;
- Branciard *et al* (2016): see-saw optimization for correlations realized in fixed-dimensional systems;
- Bavaresco *et al* (2019): constraints for GYNI correlation realized in finite-dimensional systems (trivial for infinite dimensional systems);
- Kunjwal and Oreshkov (2023): no perfect win for GYNI.

research gap: little is known about the limits of correlations with ICO.

Tsirelson bounds for correlations with ICO

The optimization problem for ICO bounds

$$\mathcal{J}^{\mathrm{ICO}} = \max_{\mathcal{S}} \max_{\left(\mathcal{M}^{(i)}_{a_i|x_i}\right)} \mathcal{S}\left(\sum_{\vec{a},\vec{x}} \alpha_{\vec{a},\vec{x}} \bigotimes_{i=1}^{N} \mathcal{M}^{(i)}_{a_i|x_i}\right)$$

Challenges

1. dimensions of quantum systems not fixed;
2. complex affine constraints for process matrices.

Explicit form of the problem

- target function: $\mathcal{J} = \sum_{\vec{a},\vec{x}} \alpha_{\vec{a},\vec{x}} p(\vec{a}|\vec{x}) = \mathrm{Tr}[S\Omega]$
- performance operator $\Omega := \sum_{\vec{a},\vec{x}} \alpha_{\vec{a},\vec{x}} \bigotimes_{i=1}^{N} M^{(i)}_{a_i|x_i}$
- contraints for **process matrices**: $S \in \mathcal{P} \cap \mathrm{DualAff}(\mathrm{NoSig})$
- contraints for **instruments**:

$$\forall i \quad \left[\left(M^{(i)}_{a_i|x_i} \right)_{a_i,x_i} \subseteq \mathcal{P}, \quad \forall x_i \quad \mathrm{Tr}_{A^{(i)}_{\mathrm{O}}} \left(\sum_{a_i} M^{(i)}_{a_i|x_i} \right) = \mathbb{I}_{A^{(i)}_{\mathrm{I}}} \right]$$

Bypass the challenges

1. For a special class of correlations, ICO bounds can be saturated by a **canonical choice of instruments**;
2. Fixing local instruments, optimization over process matrices is a **semidefinite program (SDP)**.

Single-trigger correlations

Single trigger every party i has a special setting (the trigger) ξ_i

$$x_i \neq \xi_i \implies \alpha_{\vec{a},\vec{x}} \text{ independent of } a_i$$

Example: LGYNI is single-trigger; each party's trigger is 1.

$$\Pr(x_1(a_1 \oplus x_2) = 0, x_2(a_2 \oplus x_1) = 0 \mid x_1 = 0, x_2)$$
$$= \Pr(x_2(a_2 \oplus x_1) = 0 \mid x_1 = 0, x_2)$$

Canonical instrument $\star$

- encode the setting in an auxiliary system:

$$\mathcal{M}^{(i)\star}_{a_i|x_i} = \mathcal{N}^{(i)\star}_{a_i|x_i} \otimes |x_i\rangle\langle x_i|_{\text{aux}}$$

- if $x_i = \xi_i$, measure in computational basis, otherwise do nothing:

$$\mathcal{N}^{(i)\star}_{a_i|x_i}(\rho) := \begin{cases} |a_i\rangle\langle a_i|\rho|a_i\rangle\langle a_i| & \text{if } x_i = \xi_i \\ \frac{1}{m_i}\rho & \text{otherwise (random outcome)} \end{cases}$$

- canonical performance operator $\Omega^\star := \sum_{\vec{a},\vec{x}} \alpha_{\vec{a},\vec{x}} \bigotimes_{i=1}^{N} M^{(i)\star}_{a_i|x_i}$

ICO bounds for single-trigger correlations

- in terms of **max relative entropy distance** (w.l.o.g., assume $\Omega^\star \geq 0$)

$$\mathcal{J}^{\mathrm{ICO}}_{\text{single-trigger}} = 2^{D_{\max}(\Omega^\star \,\|\, \mathrm{NoSig})}$$

- expressed through **SDP**:

$$\begin{aligned} \text{maximize} \quad & \mathrm{Tr}(S\Omega^\star) \\ \text{subject to} \quad & S \in \mathrm{DualAff}(\mathrm{NoSig}) \\ & S \geq 0 \end{aligned}$$

$$\begin{aligned} \text{minimize} \quad & \eta \\ \text{subject to} \quad & C \in \mathrm{Aff}(\mathrm{NoSig}) \\ & \eta C \geq \Omega^\star \end{aligned}$$

Sketch of the proof of single-trigger ICO bounds

$\mathcal{I}(\mathbb{M})$: the range of a correlation $\mathcal{I}$ generated by arbitrary process matrices and a restricted set $\mathbb{M}$ of instruments.

$\mathbb{M}_1 := \{\text{all instruments}\}$

$\mathbb{M}_2 := \{\text{labelled projective instruments}\}$

$\mathbb{M}_3 := \{\text{single-trigger instruments}\}$

$\mathbb{M}_4 := \{\text{the canonical instrument}\}$

Step 1 $\mathcal{I}(\mathbb{M}_1) = \mathcal{I}(\mathbb{M}_2)$;

Step 2 (single-trigger)

$\max \mathcal{I}(\mathbb{M}_2) = \max \mathcal{I}(\mathbb{M}_3)$.

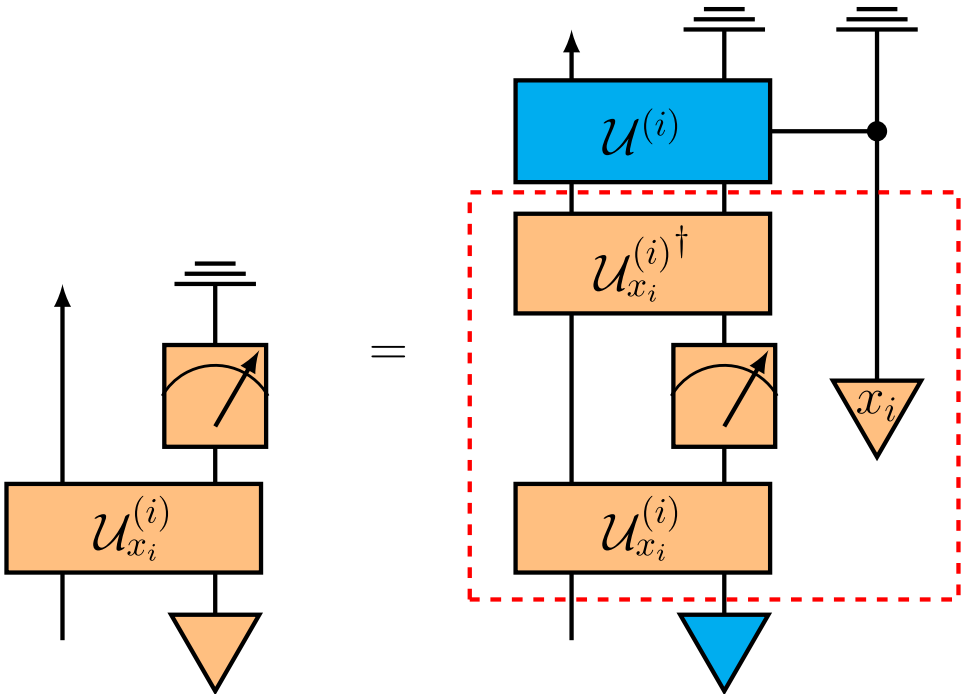
Step 3 $\mathcal{I}(\mathbb{M}_3) = \mathcal{I}(\mathbb{M}_4)$.

Step 1: reduction to labelled projective instruments

- unitary realization
- labelled projective instruments:

$$\mathcal{P}^{(i)}_{a_i|x_i} \otimes |x_i\rangle\langle x_i|_{\text{aux}}$$

where $\left(\mathcal{P}^{(i)}_{a_i|x_i}\right)_{a_i}$ is an **equal-rank** projective measurement for every setting.



Step 2: reduction to single-trigger instruments

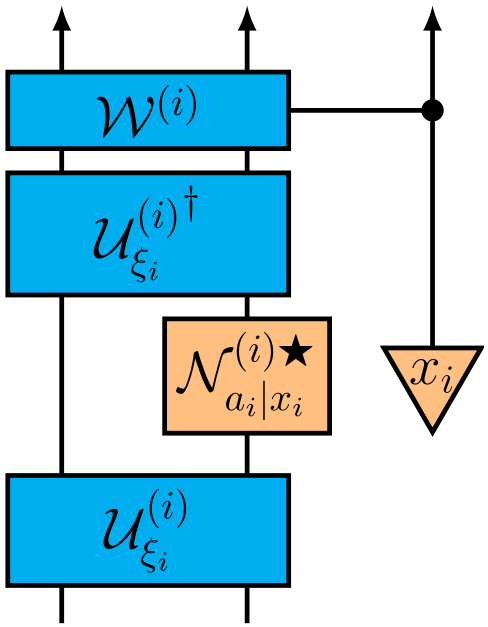
- **single-trigger instruments** (unitary for non-trigger settings):

$$\mathcal{M}^{(i)}_{a_i|x_i}(\rho) = \begin{cases} P^{(i)}_{a_i|x_i}\rho P^{(i)}_{a_i|x_i} \otimes |\xi_i\rangle\langle\xi_i|_{\text{aux}} & \text{if } x_i = \xi_i \\ \frac{1}{m_i} W^{(i)}_{x_i}\rho W^{(i)\dagger}_{x_i} \otimes |x_i\rangle\langle x_i|_{\text{aux}} & \text{otherwise} \end{cases}$$

- the value of a single-trigger correlation is unchanged when **discarding the outcomes** for non-trigger settings;
- a projective measurement followed by discarding the outcome is equivalent to a **random unitary channel**.

Step 3: reduction to the canonical instrument

- Single-trigger instruments can be obtained from the canonical instrument locally.



Single-trigger SDP relaxation

SDP relaxation

$$\mathcal{J} \leq \min_{\{\mathcal{J}_j\}} \left\{ \sum_j \mathcal{J}_j^{\mathrm{ICO}} \right\}$$

- $n_1 n_2 \cdots n_N$ choices for triggers;
- decompose $\mathcal{J}$ into a **sum of single-trigger correlations** $\{\mathcal{J}_j\}$;
- at most $n_1 n_2 \cdots n_N$ non-zero terms in $\{\mathcal{J}_j\}$.

ICO bounds for LGYNI and OCB

$$\mathcal{J}^{\mathrm{ICO}}_{\mathrm{LGYNI}} = 0.8194, \qquad \mathcal{J}^{\mathrm{ICO}}_{\mathrm{OCB}} = 1 + \frac{1}{\sqrt{2}}$$

OCB correlation written as a random mixture of single-trigger ones:

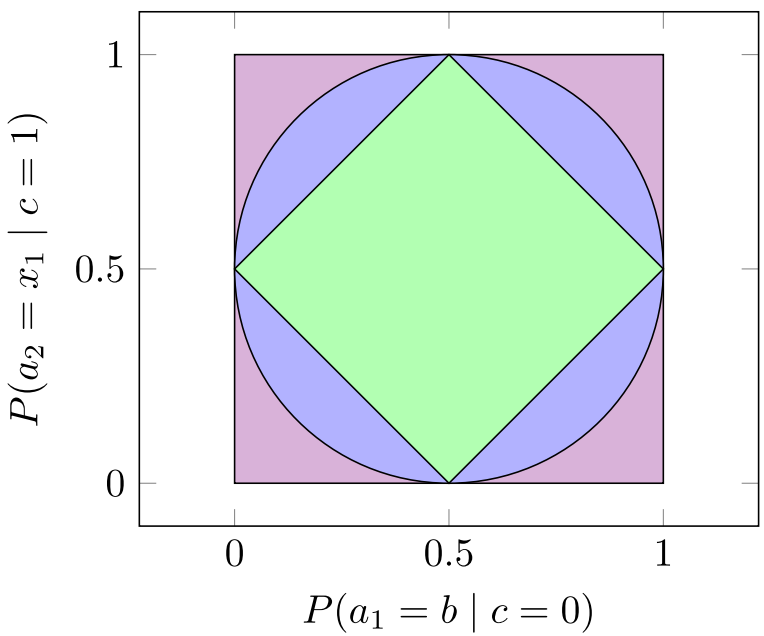
$$\mathcal{J}_{\mathrm{OCB}} = \tfrac{1}{2}[\Pr(a_1 = b | x_1 = 0, c = 0) + \Pr(a_2 = x_1 | b = 0, c = 1)]$$
$$+\tfrac{1}{2}[\Pr(a_1 = b | x_1 = 1, c = 0) + \Pr(a_2 = x_1 | b = 1, c = 1)]$$

An upper bound for GYNI

$$\mathcal{J}_{\text{GYNI}}^{\text{ICO}} \leq 0.7592$$

$$\begin{aligned}
\mathcal{J}_{\text{GYNI}} \leq\ & \alpha_0 p(00|00) + \alpha_1 p_A(1|01) + \alpha_2 p_B(1|10) \text{ triggers } (0,0) \\
& + \alpha_0 p(10|01) + \alpha_1 p_A(0|00) + \alpha_2 p_B(1|11) \text{ triggers } (0,1) \\
& + \alpha_0 p(01|10) + \alpha_1 p_A(1|11) + \alpha_2 p_B(0|00) \text{ triggers } (1,0) \\
& + \alpha_0 p(11|11) + \alpha_1 p_A(0|10) + \alpha_2 p_B(0|01) \text{ triggers } (1,1)
\end{aligned}$$

A slice of the set of probability distributions with ICO



- ICO bound for $\mathcal{J}_{\mathrm{OCB},\alpha}$:

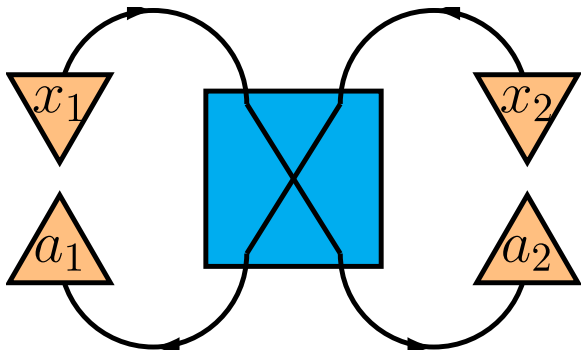
$$\frac{1}{2}\left(1 + \alpha + \sqrt{1 + \alpha^2}\right)$$

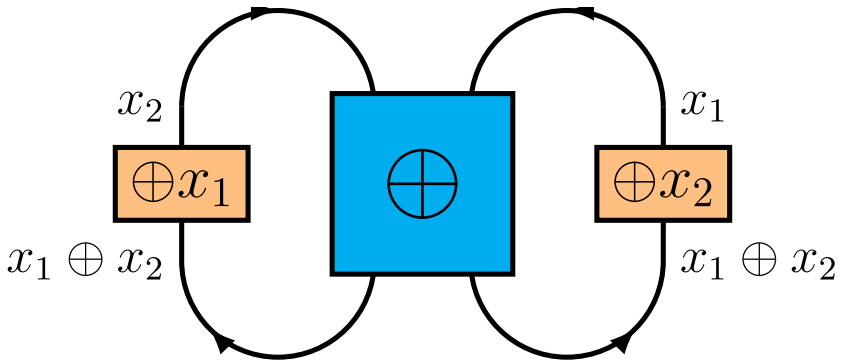
- blue circle: ICO set, **NOT** a polytope
- green square: causal polytope

$$\mathcal{J}_{\mathrm{OCB},\alpha} := \Pr(a_1 = b \mid c = 0) + \alpha \Pr(a_2 = x_1 \mid c = 1)$$

Go beyond ICO bounds

$$p(a_1, a_2 \mid x_1, x_2) = \delta_{a_1,x_2}\delta_{a_2,x_1}$$





R. Kunjwal and O. Oreshkov (2023)

G. Chiribella and Z. Liu (2022)

Open questions and outlook

Open questions

- Is the single-trigger SDP relaxation tight for correlations other than single-trigger ones and OCB correlation?
- Is our bound for GYNI tight?
- Is there a hierarchy of SDP relaxations for the ICO-bound problem?

Outlook

- bounds for quantum correlations with partially pre-defined causal structure, such as those violating DRF inequalities;

T. van der Lugt, J. Barrett, G. Chiribella (2023)

- device-independent protocols, such as self-testing, randomness amplification, and DIQKD;
- principles for ICO, such as non-trivial communication complexity, non-trivial nonlocal computation, information causality, macroscopic locality, and local orthogonality.

Thank you for your attention!